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Media Intelligence Report

Analyzing the evolution of media perception of X (Formerly Twitter)

April 2026

Contents

Context and Problem Definition

X, formerly known as Twitter, is one of the most significant social media platforms of the past twenty years and has emerged as a vital case study in technology, governance, and reputation management. Following Elon Musk's $44 billion takeover in October 2022, the company has experienced a significant and apparent transformation: extensive layoffs, complete rebranding, departure of advertisers, and notable controversies around content moderation. The choice to concentrate on X for this research is driven by its significant public attention and a personal interest in comprehending the interplay between media perception and corporate decision-making.

Media perception is a measurable business signal. Negative media coverage diminishes advertiser confidence, hastens user attrition, reduces valuation, and attracts regulatory examination. For a platform reliant on advertiser confidence and user interaction, its media representation is not mere background noise; it directly influences commercial results. This paper analyses perceptions from October 2022 to January 2026, using coverage from two prominent broadsheets, The Guardian and The New York Times. The two-source approach was intentional. An initial analysis of Guardian only revealed a predominantly negative bias, and to ascertain if this pattern represented genuine perception or single-outlet bias, a second source with a different editorial value was used.

Data Extraction and Preparation

Articles were extracted using two REST APIs: The Guardian open Platform and The New York Times Article Search API. Both return JSON, which was flattened into pandas Data Frames utilizing a common schema (id, date, section, abstract or trail text, body, URL). A three-tired filter was subsequently implemented; a section limitation at the API level (Technology, Business, media for the Guardian; Business, Technology , Opinion for the New York Times); a Boolean Keyword query integrating a platform anchor with themes of crisis or strategic shift; and a regex check after fetching to confirm the headline or trail text was genuinely about X or Twitter rather than just mentioning it in passing. This was done to remove noise from the dataset and ensure only relevant articles were retained for analysis. The pipeline underwent continuous refinement. The initial Guardian query inadequately represented the acquisition period from October 2022 to February 2023, as the changing terminology of a developing crisis did not align with the established themes; the regex was expanded to include post-rebrand terminology

(“Musk’s platform”, “X users”, “Yaccarino”). Upon completion of the cleaning process, the final corpus comprised 295 articles from the Guardian and 197 articles from the New York Times, resulting in total of 492 articles across 41 months.

Comparison of Approaches and Selection

Two analytical approaches were considered. The Conventional NLP workflow combines tokenization, lexicon-based sentiment scoring (e.g., VADER), TF-IDF keyword analysis and topic modeling via LDA. The Generative AI workflow sends each article to a large language model with a structured prompt and receives back classified outputs (sentiment band, theme, driver event, rationale) as JSON.

Dimension	NLP	Gen AI
Depth of Insight	Sentiment is number; topics are word clusters needing manual labelling.	Returns sentiment, named themes, specific drivers and a one-line rationale per article.
Accuracy & reliability	Reliable for straight forward positive and negative classification.	More accurate on long-form journalism; interprets tone and sarcasm.
Scalability & interpretability	Free, fast, fully auditable.	Par-call cost; model’s reasoning is partially opaque.
Business usefulness	Useful baseline; outputs need translation for executives.	Outputs map directly to executive vocabulary (themes, drivers, stance).

Table 1 Comparison of NLP and GenAI workflows across four dimensions.

The Generative AI workflow was selected. Deciding criteria were the features of the corpus and the nature of the research question. The dataset (492 articles) is small enough that the cost objection to Gen AI is negligible. Guardian articles on average roughly 600 words and frequently contain irony, subtle claims, and cited sources, characteristics of long-form texts that yield imprecise and potentially deceptive averages when assessed by lexicon-based scoring produces noisy, often misleading averages. Most importantly, the assignment asks for drivers of perception change, which GenAI returns as structured field; with NLP this would require running topic modelling, manually label clusters, and writing rules to map each article to driver, substantial engineering for an inferior outcome. GenAI excels in contextual nuance with lengthy texts, whereas traditional NLP is more effective with brief, consistent inputs like product reviews. (Zuliani, 2026).

The accepted trade-off is reproducibility. Despite a temperature setting of zero, GenAI outputs remain imperfectly predictable, exemplifying the “black box” phenomenon. This was mitigated by persisting per-article classifications to CSV (so downstream analysis is fully reproducible from that file) and by manually validating a stratified sample of outputs.

Applied Data Analysis

A two-pass GenAI architecture was implemented using Gemini 2.5 Flash. Pass 1 classified each article individually for sentiment (5-point scale), primary theme (12-category fixed list), driver event (free text, max 5 words), and stance toward the company. Pass 2 aggregated those classifications into monthly summary tables and asked the model to synthesize trajectory and drivers, with explicit instructions to use only the supplied data.

Sentiment evolution

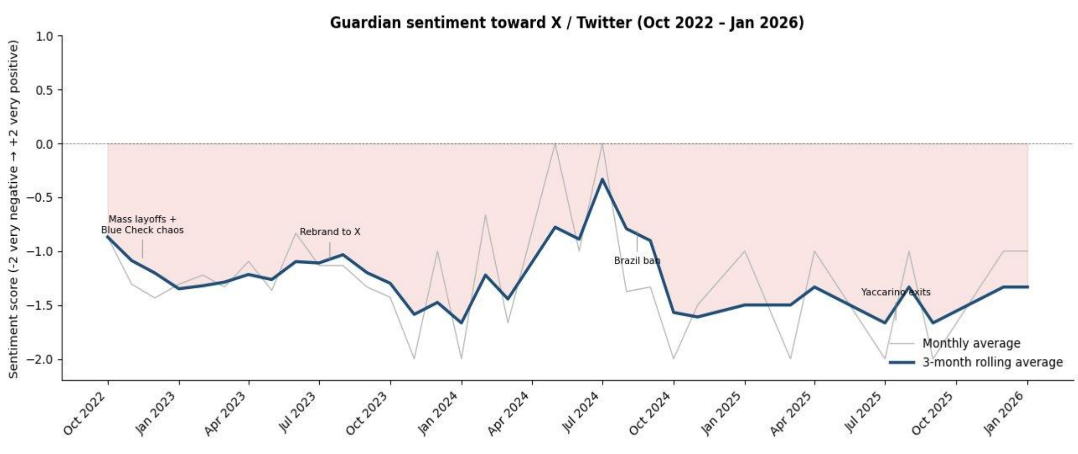

Figure 1. Guardian monthly sentiment toward X (Oct 2022 – Jan 2026), 3-month rolling average

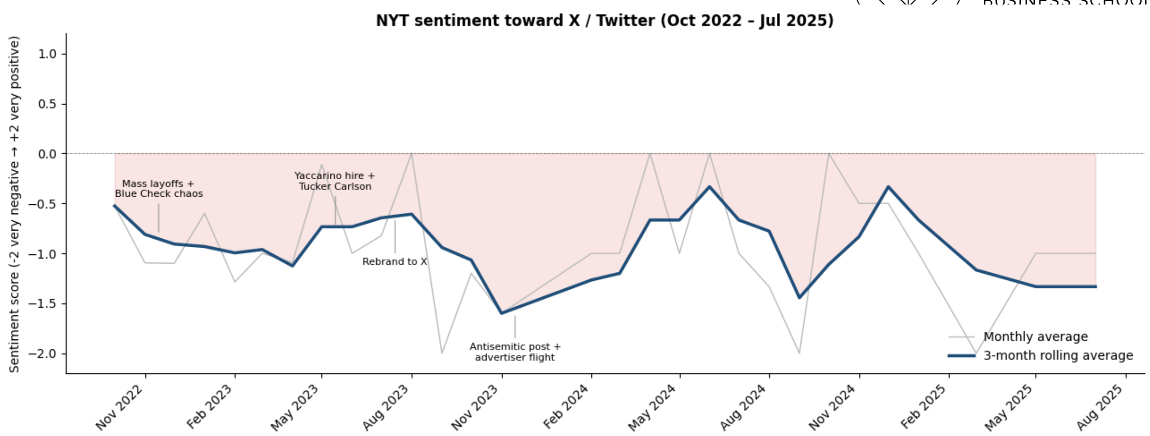

Figure 2. NYT monthly sentiment toward X (Nov 2022 – Aug 2025), 3-month rolling average

Both outlets show sentiment never crossing into positive territory. Guardian coverage moved through three phases, Acute crisis (Oct 2022-Feb 2023 at 1.4), rebrand transition (mid 2023), and post rebrand floor of -1.3 to -1.6. NYT followed a similar arc but less sharply, briefly recovering around the Yaccarino hire and rebrand before collapsing in November 2023 after Musk's antisemitic post and the advertiser exodus. In both, recovery proved structurally elusive even strategic resets like the rebrand failed to break the negative trajectory.

Sentiment composition

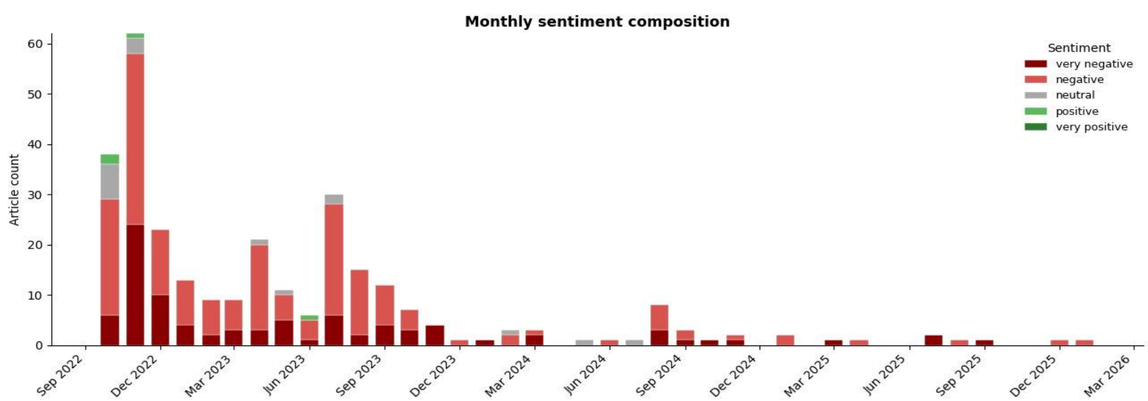

Figure 3. Guardian monthly sentiment composition. Positive coverage (green) is virtually absent.

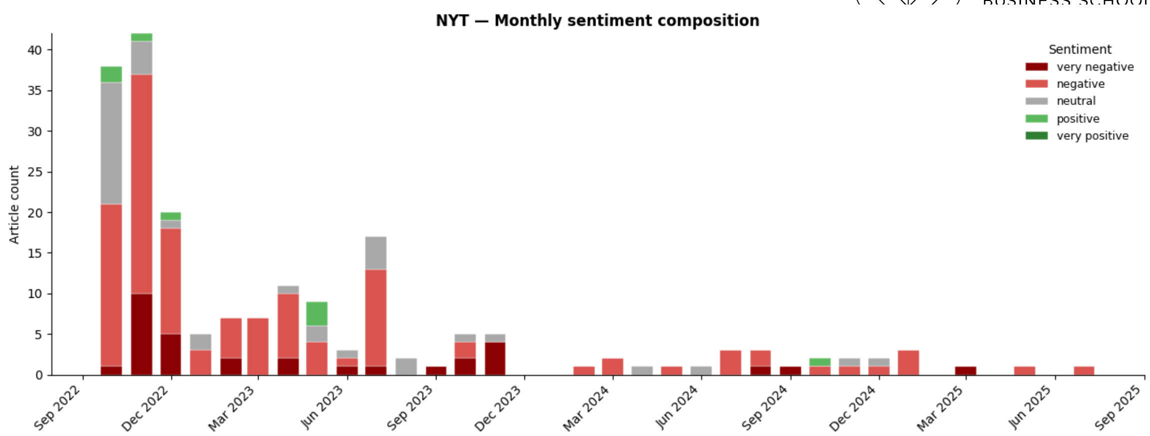

Figure 4. NYT monthly sentiment composition.

Of 295 Guardian articles, only 4 (1%) were classified as positive. Closer manual inspection revealed that three of these were not positive about X under Musk at all but retrospectives of pre-acquisition Twitter. NYT showed a slightly higher positive count (8 articles, 4%), mostly clustered in late 2022 and mid 2023 around the rebrand and CEO appointment, but these were similarly isolated and short-lived. Across both outlets and 41 months, neither produced a sustained positive narrative about X under Musk's ownership.

Themes and drivers

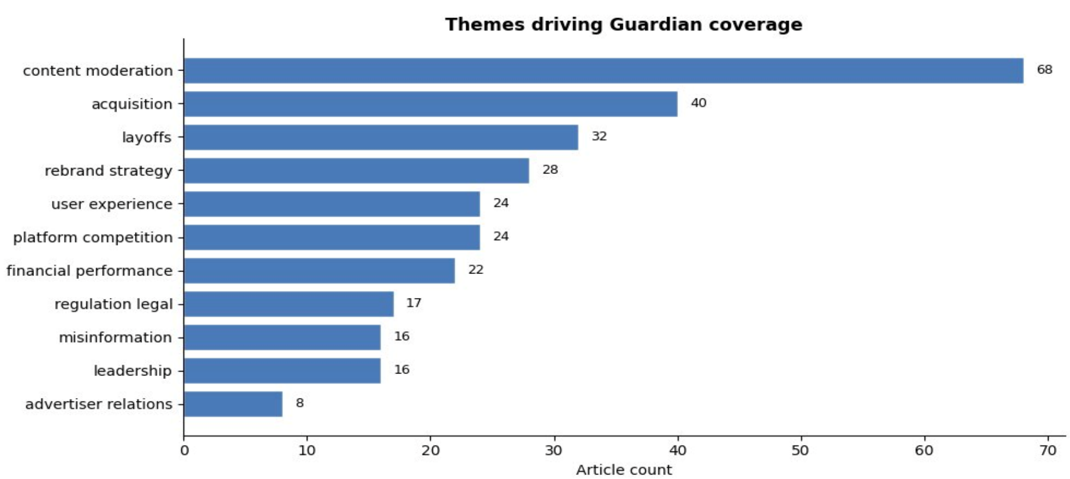

Figure 5. Theme distribution in Guardian coverage. Content moderation dominates.

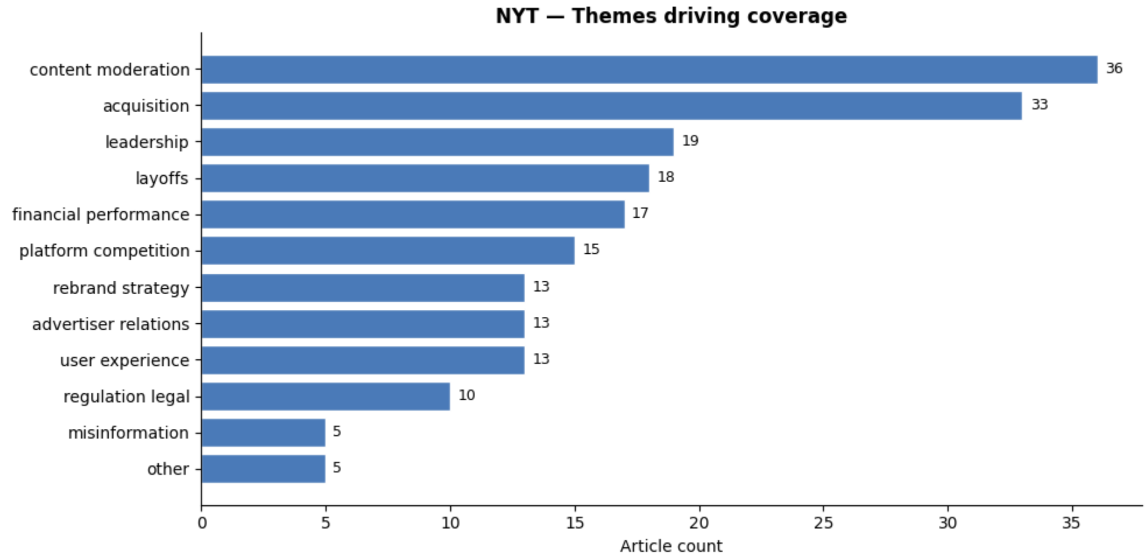

Figure 6. Theme distribution in NYT coverage. Content moderation dominates.

Both outlets share content moderation and acquisition as the top two themes, but NYT places greater weight on leadership and financial performance, reflecting a more procedural US business-press lens.

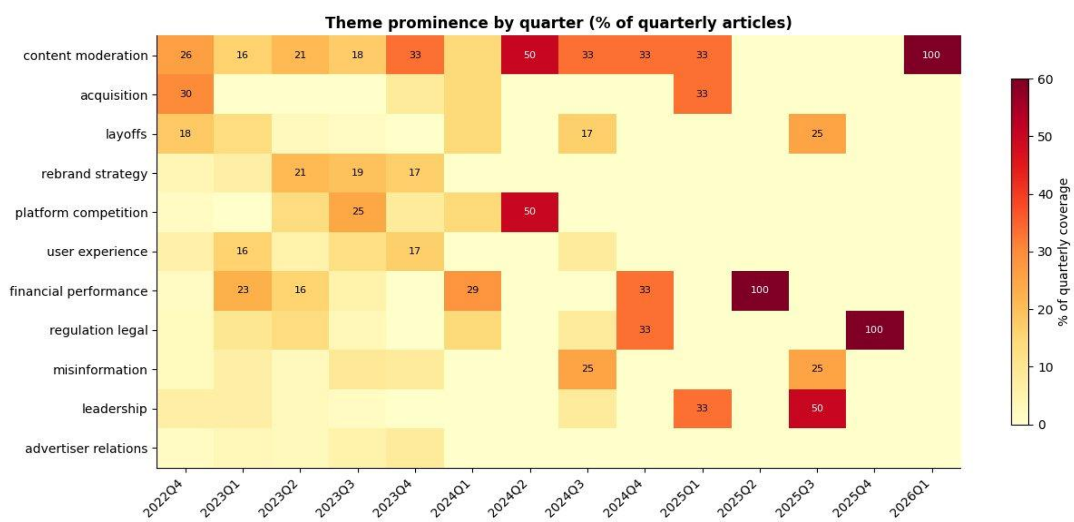

Figure 7. Theme prominence by quarter. The story arc is visible: acquisition → layoffs → moderation → rebrand → competition → regulation.

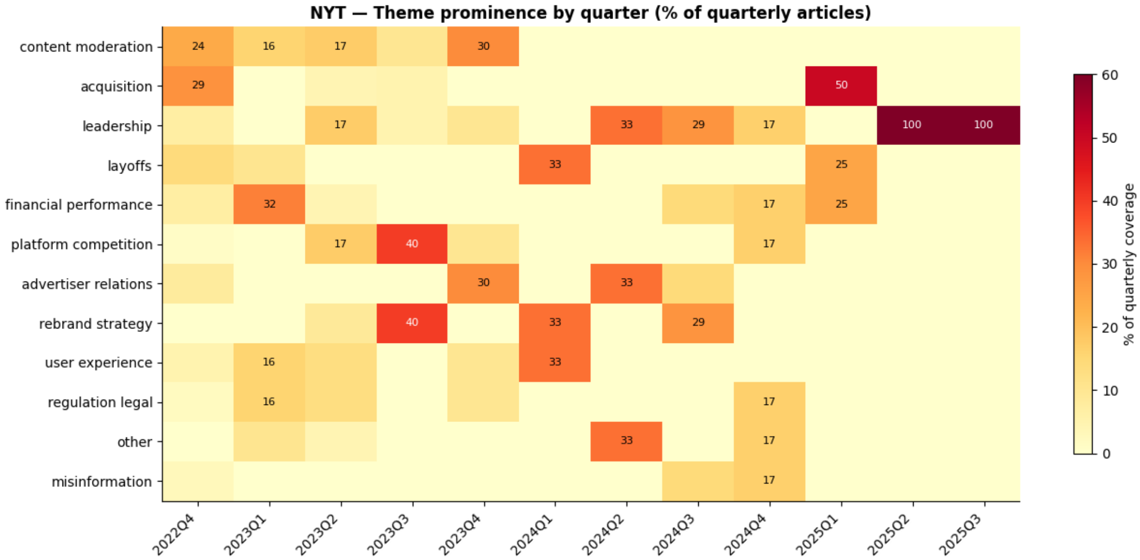

Figure 8. NYT theme prominence by quarter (% of quarterly articles). Late-period 100% values (2025 Q2, Q3) reflect single-article quarters rather than editorial dominance.

The heatmap shows a cascading series of editorial frames rather than single narrative. NYT follows the same arc acquisition and moderation early, rebrand and competition in mid-2023, leadership from 2024. Two outlets producing the same thematic progression independently strengthens the finding: addressing any one theme alone would not shift the overall picture.

Cross-source validation

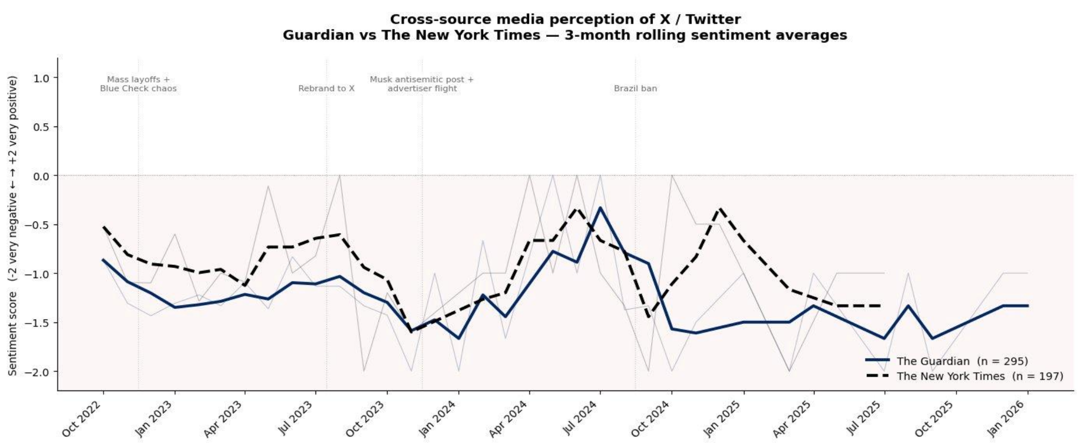

Figure 9. Cross-source comparison: Guardian (n=295) vs The New York Times (n=197), 3-month rolling sentiment.

The same process was utilized to examine single-source bias. Both outlets consistently register below zero for the entire 33-month overlap. Their monthly trajectories exhibit a correlation (Pearson r = 0.53, n = 26 monthly observations), with both demonstrating concurrent negative inflections in response to identical external events: the layoffs in

November 2022, the antisemitic post in November 2023, and the subsequent advertiser exodus. The New York Times routinely outperforms The Guardian, with a mean sentiment score of -0.89 compared to the Guardian's -1.22, and 16% of New York Times articles classified as very negative, in contrast to 31% for the Guardian. The convergence validates the existence of a perception crisis: the divergence suggests that intensity is partially contingent upon editorial choices (the Guardian presents the narrative more acutely, while the NYT employs a more procedural lens typical of US business journalism).

Insights and Interpretation

Three perspectives emerge from the analysis. First, X's reputational damage is structural rather than episodic; perception did not improve following any significant event in either dataset, including intentional resets such as the rebranding. Secondly, the number and intensity of coverage have diverged: the frequency of articles dropped significantly post- 2023 (from 62 Guardian stories in November 2022 to single-digit monthly totals by mid-2024), yet when coverage is present, it consistently maintains a critical stance. This decoupling serves as a diagnostic indicator diminished coverage signifies not recovery, but rather a decrease in editorial significance. Third, drivers distinctly separate into internal decisions (Musk's pronouncements, layoffs, moderation strategies) and external pressures (regulatory measures, advertising actives, competition introductions), with internal factors frequently aligning with the most pronounced negative fluctuations. Numerous constraints must be recognized. The dataset is confined to two anglophone broadsheets: including newspapers (FT, WSJ), and non-English sources would enhance the analysis. Beginning in 2024, monthly sample numbers will decrease to single digits in both sources, rendering month-level interpretation less credible; thus, 3- month rolling averages are employed in Figures 1 and 5. The GenAI classifier exhibits inherent biases in its training data and sometimes over-classifies marginally positive framings; a manual inspection of four "positive" articles from the Guardian revealed one borderline instance. The significant findings remain intact after this adjustment.

Conclusion and Recommendations

Media perception of X has evolved through a consistently negative trajectory shaped by identifiable events. Both Guardian and NYT independently produced overwhelmingly negative coverage (93% and 77% negative or worse respectively). This convergence rules out single-outlet bias as the explanation. Outlet-specific intensity differs but directional consensus does not. The reputational position is structural and multidimensional, with content moderation and leadership conduct as the most persistent recurring drivers.

Three Strategic recommendations follow:

- **Treat reputation recovery as a multi-front programme, not a single campaign.** The thematic heatmap shows negative coverage cycles through different editorial frames (Moderation, rebrand, regulation, layoffs). Addressing only one will not move the aggregate.
- **Manage leadership conduct as a corporate risk.** The most significant negative changes in both outlets coincided with Musk's controversial public statements. Regard CEO communications as systematic corporate risk, rather than mere personal expression, to reduce the primary volatile factor influencing impression.
- **Use the decoupling of volume and intensity as an early-warning signal.** Reduced coverage frequency is being misread internally as stabilization; the data indicates the opposite. Coverage has grown less frequent but uniformly critical. A monitoring dashboard tracking both volume and tone would give leadership a more honest read on reputational health.

More broadly, this report demonstrates the value of combining structured GenAI classification with multisource validation. The convergent evidence across Guardian and NYT, neither of which the company can dismiss as fringe, provides a defensible empirical basis for decisions that could otherwise depend on impression and anecdote.

References

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Appendix 1

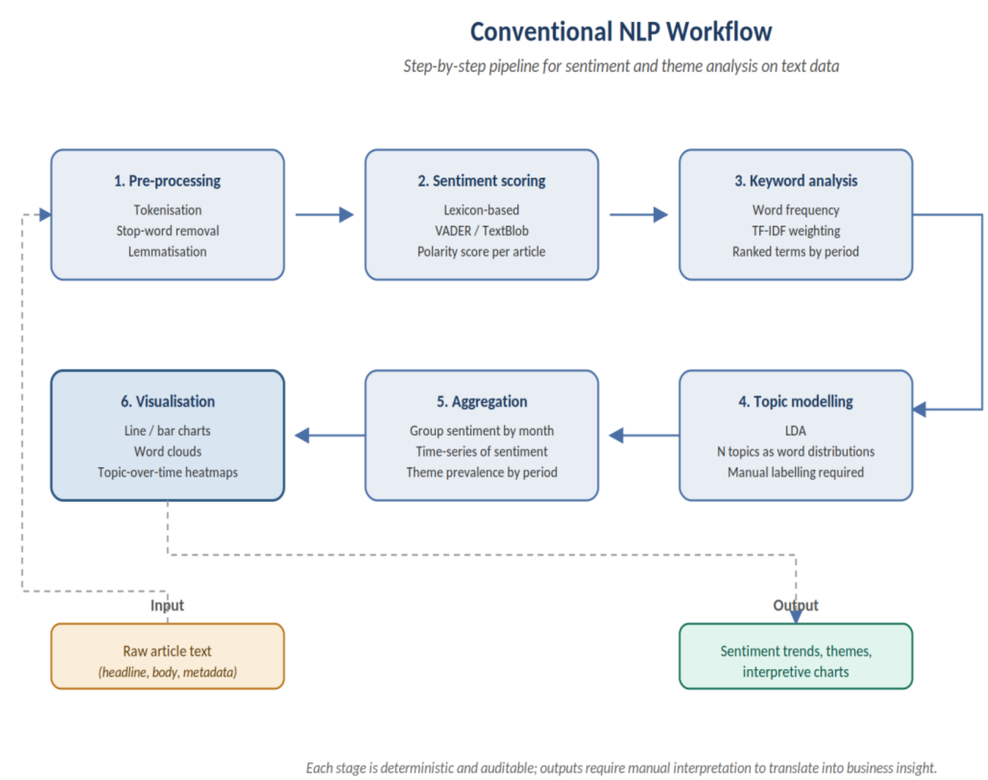

Figure 10 Conventional NLP Workflow

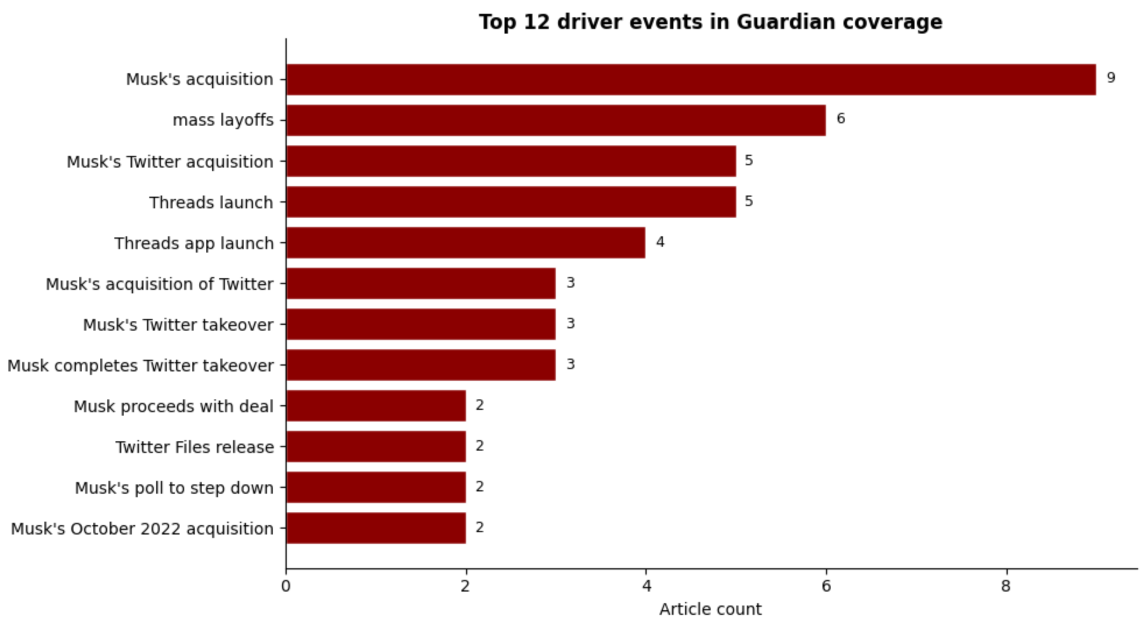

Figure 11 Top Drivers of Events Guardian

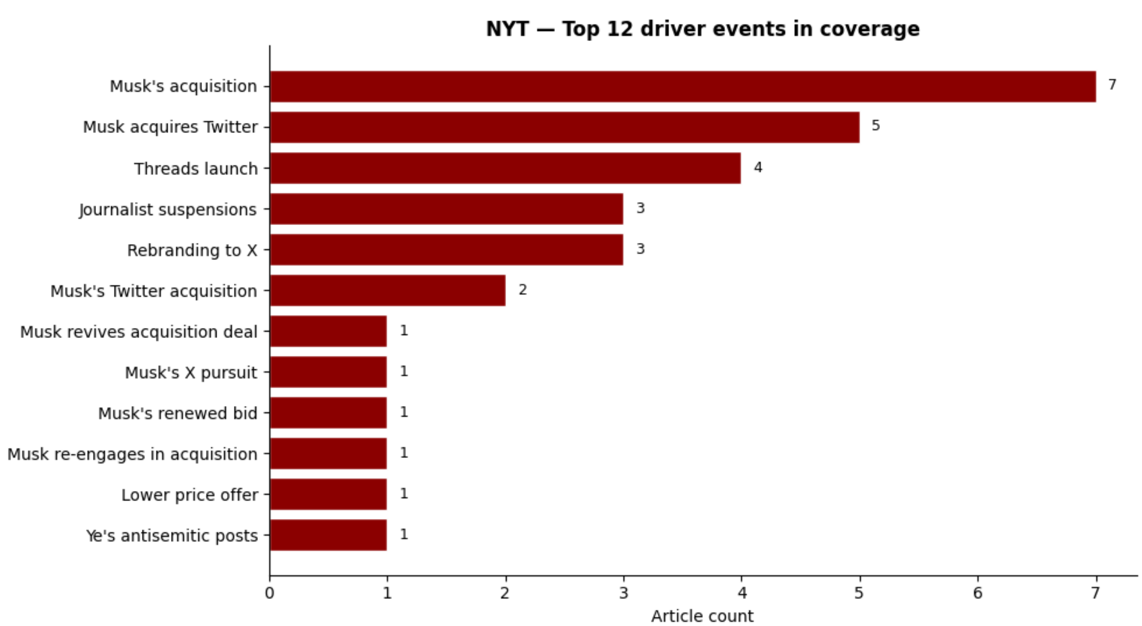

Figure 12 Top Drivers of Events NYT

Appendix 2

- **AI prompts used for Coding**

    - Rewrite the code to select articles that has X, twitter or elon musk in it
    - This code gave alot of unnecessary news, which is noise. suggest words that will tailor it specifically to the controversy, crisis, or reputational issue, OR a major strategic shift that has happened in 'X' or former twitter.
    - I still have noise, the Australian news. etc. .... Pls write a new code, I want to start again. including all the checks and adjustments to filter out the noise. Write it from all the imports.
    - Write a code for the below
      Check that data was successfully retrieved
      Select the first article from the list
      Formats the JSON so it is easier to read
      Displays the raw structure of one article

- Write a code that checks that articles were successfully retrieved
  Creates a Dataframe from a list of dictionaries
  Extracts only the date portion from the full timestamp
  Keeps the article headline
  Keeps the article link
  Shows the first 10 rows so we preview the table
- Before analyzing time-based patterns, dates must converted into a format python can recognize and work with . The original date column in a proper date format , a new month column with values .

- Count the number of articles by month Now that the data is properly structured and the date field has been prepared, we can begin analysing patterns. In this step, we count how many articles were published each month. This allows us to track how media attention on the topic evolves over time.
- What this __code__ Download codeis doing

- df.groupby("month") → groups all articles by month
- .size() → counts how many articles fall into each month

- .reset_index(name="article_count") → converts the result into a table with a named column
- monthly_counts["month"].astype(str) → converts the month into a readable string format for plotting

We are transforming raw unstructured data into a structured measurable business signal — media attention over time.

This type of analysis is commonly used to:

- monitor emerging topics
- track public interest
- identify spikes in attention

However, it is still not very easy see the trends. This is where data visualisation comes in.

- I want to start writing section 4 code I will be using Google Gemini API. I already have the API key, i just need to, before running any analysis you need to authenticate with your API [key. so, write me the code i will use to connect the gemini to my guardian Json file I already have in my google colab.

- **Full prompts used in the GenAI analysis (Both prompts were used for Guardian and NYT)**

- How has the media perception of ''X'' or Formally known as 'Twitter' evolved over time, and what are the key drivers behind this change? 1. How has the overall sentiment (positive, negative, neutral) toward the organization changed over time (e.g., monthly or yearly trends)?

Are there any significant spikes or drops in sentiment, and what time periods do they occur in?

. Provide a short executive summary (maximum 120 words) explaining what these suggest about the company Return your answer in clear bullet points under each heading.
write me the code for this
i had this prompt can it b woven into what you already wrote above

- **Output**

==
EXECUTIVE ANALYSIS
==

1. Overall sentiment evolution over time
* The initial period following Elon Musk's acquisition in October 2022 saw a rapid decline in media perception, with average sentiment dropping from -0.87 (38 articles) to a low of -1.43 by December 2022 (23 articles), driven by themes of layoffs and content moderation. This marks an acute crisis phase.
* Throughout 2023, sentiment remained predominantly negative, fluctuating between -0.83 (June 2023, 6 articles) and -1.43 (October 2023, 7 articles), with the rebrand to X and the launch of Threads contributing to a dip in July 2023 (-1.13, 30 articles). This represents a turbulent transition phase.
* From late 2023 into 2026, coverage became significantly sparser, often featuring only 1-4 articles per month, yet sentiment frequently plunged to its lowest possible score of -2.00 (e.g., November 2023, January 2024, March 2025, July 2025, September 2025, October 2024).
* Despite the overall negative trend, there were isolated instances of neutral sentiment (May 2024, July 2024) and the least negative average sentiment of -0.67 in February 2024, though these occurred during periods of extremely low article volume (1-3 articles).
* This trajectory suggests an initial acute crisis, followed by a turbulent transition phase, and then a period of sustained, often intensely negative, but less frequent coverage, indicating a deeply entrenched negative reputational position.

2. Significant spikes and drops in sentiment
* **November 2022** saw a significant drop in sentiment from -0.87 to -1.31 (a drop of 0.44). This decline was largely driven by `mass layoffs` and the ongoing fallout from `Musk's acquisition of Twitter`, highlighting initial operational turmoil and negative public perception around `layoffs` and `content_moderation`.

*   A notable improvement occurred in **June 2023**, with sentiment rising from -1.36 to -0.83 (an increase of 0.53). This shift, despite drivers like `Twitter quits EU voluntary code` and `Meta's internal meeting on rival app`, suggests a temporary stabilization or a perception of decisive action in `regulation_legal` and `platform_competition`.
*   The sentiment plummeted again in **July 2023**, dropping from -0.83 to -1.13 (a drop of 0.30). This was primarily due to the `Threads launch` and `Twitter rebrands to X`, indicating negative reception to the `rebrand_strategy` and increased `platform_competition`.
*   The sharpest decline in sentiment occurred in **November 2023**, falling from -1.43 to -2.00 (a drop of 0.57). This was directly linked to `Musk endorses antisemitic tweet` and `Musk's antisemitic tweets`, severely impacting `content_moderation` and `advertiser_relations`.
*   A significant, albeit short-lived, sentiment spike was observed in **February 2024**, improving from -2.00 to -0.67 (an increase of 1.33). While drivers like `Musk ordered to testify` and `ASA ruling on CBD tweets` are regulatory in nature, the relative improvement suggests a temporary shift away from the intense personal controversies of the preceding months, focusing on `regulation_legal` and `acquisition` themes.

3. Executive summary
The Guardian's coverage of X since October 2022 reveals a consistently negative media perception, evolving from an initial acute crisis post-acquisition to a period of sustained, often intensely negative, but lower-volume reporting. Key drivers include widespread layoffs, controversial content moderation decisions, significant leadership controversies, and challenges from platform competition like Threads. The rebrand to X also coincided with a sentiment drop. While there were brief, minor improvements, these were often short-lived and occurred amidst very low article counts. This trajectory suggests X faces a deeply entrenched reputational challenge, struggling to stabilize its platform, retain advertisers, and manage public perception amidst ongoing strategic and operational turbulence.

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Prompt 2

Write a code for this

Drivers of Perception Change

- What specific events, announcements, or incidents coincide with major changes in sentiment?
- Which keywords, entities (e.g., executives, regulators, competitors), or phrases frequently appear during sentiment shifts?

- Are there any external factors (e.g., economic conditions, regulation changes, industry crises) influencing media perception?
- 2. . Provide a short executive summary (maximum 120 words) explaining what these suggest about the company Return your answer in clear bullet points under each heading.

Output

/tmp/ipykernel_763/176780715.py:22: UserWarning: Converting to PeriodArray/Index representation will drop timezone information.

results["month"] = results["date"].dt.to_period("M").astype(str)

==

DRIVERS OF PERCEPTION CHANGE — ANALYSIS

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1. Specific events, announcements, or incidents coinciding with major sentiment changes

* **November 2022:** A sentiment change of -0.44 (from -0.87 to -1.31) was driven by `mass layoffs` and `Musk's acquisition of Twitter`. The widespread job cuts immediately following Musk's takeover likely fueled negative media perception regarding the company's stability and ethical practices.

* **July 2023:** A sentiment change of -0.30 (from -0.83 to -1.13) coincided with `Threads launch`, `Twitter rebrands to X`, and `tweet viewing limits imposed`. The introduction of a direct competitor (Threads), coupled with the controversial rebrand and unpopular user restrictions, significantly contributed to negative sentiment.

* **November 2023:** A sharp sentiment decline of -0.57 (from -1.43 to -2.00) was primarily driven by `Musk endorses antisemitic tweet` and `Musk's antisemitic tweets`. Elon Musk's personal endorsement of antisemitic content directly led to a severe drop in media perception, highlighting the impact of his controversial statements.

* **May 2024:** A significant positive sentiment change of +1.67 (from -1.67 to 0.00) was observed when `Dorsey quits Bluesky board`. The departure of Jack Dorsey from the board of a rival platform, Bluesky, likely provided a perceived boost for X, interpreted as a weakening of a competitor.

* **August 2024:** A substantial sentiment drop of -1.38 (from 0.00 to -1.38) was driven by `Don Lemon sues X`, `Musk's tweets promoting conspiracy theories`, and the `platform used for unrest` during `UK riots after Southport stabbing`. Multiple negative events, including a high-

profile lawsuit, Musk's controversial online activity, and the platform's association with real-world unrest, collectively caused a sharp decline.

* **July 2025:** A sentiment change of -1.00 (from -1.00 to -2.00) was linked to `Linda Yaccarino stepping down` and `Musk's 'Go fuck yourselves' comment`. The departure of the CEO and Musk's highly provocative public statement likely contributed to a significant drop in sentiment, indicating leadership instability and controversial communication.

2. Recurring entities, themes, or phrases visible during sentiment shifts

* **Elon Musk's personal actions and leadership:** Elon Musk is a central and recurring driver of negative sentiment, with events such as `Musk's poll to step down`, `Musk's Super Bowl tweet flop`, `Musk endorses antisemitic tweet`, `Musk's anti-DEI actions`, and `Musk's 'Go fuck yourselves' comment` consistently impacting perception.

* **Content Moderation and Misinformation:** The themes of `content_moderation` and `misinformation` frequently appear during negative shifts, driven by events like the `Twitter Files release`, `re-allowing political ads`, `Sellner's account restored`, and `Grok AI misinformation`, indicating ongoing concerns about platform safety and integrity.

* **Layoffs and Legal Disputes:** `Layoffs` (e.g., `mass layoffs`, `executives sue for severance`) and various `regulation_legal` issues (e.g., `gender discrimination lawsuit`, `lawsuit over non-payment`, `Don Lemon sues X`, `EU fine for DSA breaches`) consistently contribute to negative media perception, highlighting operational instability and legal challenges.

* **Financial Performance and Advertiser Relations:** Concerns around `financial_performance` (e.g., `advertisers paused spending`, `Fidelity marks down X value`) and `advertiser_relations` are recurring themes, reflecting ongoing challenges in X's business model and brand safety for advertisers.

* **Leadership Instability and User Experience:** Changes in `leadership` (e.g., `Linda Yaccarino appointed CEO`, `Linda Yaccarino stepping down`) and issues with `user_experience` (e.g., `platform outage`, `tweet viewing limits imposed`, `stripping headlines from news links`) frequently coincide with negative sentiment, pointing to internal operational and strategic challenges.

3. External factors influencing perception

* **Competitor Activity:** The launch of rival platforms, notably `Threads launch` and `Threads app launch`, significantly impacted X's perception by creating competitive pressure. The event `Dorsey quits Bluesky board` also reflects external competitor dynamics influencing X's standing.

* **Regulatory and Legal Scrutiny:** External regulatory bodies and legal challenges consistently influence X's perception. Examples include `Twitter quits EU voluntary code`, `DoJ

alleges privacy order violations`, `SEC investigation revealed`, `ASA ruling on CBD tweets`, and `EU fine for DSA breaches`, highlighting ongoing compliance pressures from external authorities.

*   **Advertiser Actions:** Decisions by advertisers, such as `advertisers paused spending` and `ABC exits X`, represent significant external financial pressure and directly reflect concerns from external businesses about X's brand safety and content environment.

*   **Intersection with Real-World Events:** X's role in broader societal and political events, such as `UK riots after Southport stabbing` where the `platform used for unrest`, and `Trump's election victory`, demonstrates how external events can draw scrutiny to the platform's policies and impact.

4. Executive summary

Media perception of X is predominantly negative, driven by a combination of internal decisions and external pressures. Elon Musk's controversial personal actions and statements, alongside significant internal operational changes like mass layoffs, content moderation shifts, and user experience alterations, are primary internal drivers of sentiment decline. Concurrently, external factors such as competitor launches (Threads), increasing regulatory scrutiny (EU fines, DoJ investigations), and advertiser withdrawals significantly exacerbate negative perceptions. The platform's entanglement with real-world events further complicates its image. Strategically, X faces a persistent challenge in stabilizing its brand, requiring a concerted effort to mitigate the impact of leadership controversies, address core platform issues, and proactively manage external regulatory and competitive landscapes.

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